Module 1 — Read 0

Project preview — Project 1 — Testing a Claim

Module 1, Read 0: Why this project matters

Big question: Is what we observed **surprising**?

Project: Project 1 — Testing a Claim · **Sample report:** ESP binomial test (PDF) · **Next:** Read 1

ℹ Short read — start here

This page is a **five-minute orientation**, not the textbook. Read it once before Read 1 so you know where Module 1 is heading.

The story in one pass

Imagine you watch a repeatable process many times. You count how often something "succeeds." After $n = 25$ trials you have x successes and an observed proportion $\hat{p} = x/n$.

That number lives in the **world we observe** — real data from your study.

Before you can say whether x is unusual, you enter the **world we imagine**. You pick a value for p (probability of success on one trial) and **pretend** that value is true. That assumption defines a **binomial model**: if H_0 were correct, what counts would be typical?

The **p-value** answers: *If my imagined model were true, how often would I see something at least this extreme?* A small p-value means your observed x would be rare under that model — the data and the claim in H_0 do not fit well.

You will not need every formula on this page yet. The point is to see the **arc** of Project 1 before the readings and labs fill in the vocabulary.

Tour the sample project

Open the **sample one-page write-up — ESP binomial test (PDF)**. It is a finished Project 1 report for an ESP study. **Do not copy ESP as your topic** — it is a class example. Use the PDF to see **shape and tone**, not your research question.

Read the sample with these questions in mind:

Section	What to notice	Why it matters for *your* project
Introduction	Research question, motivation, parameter p, H_0 and H_a with a numeric null	You will state these **before** collecting $n = 25$ trials (milestone hub · notebook)
Data collection	How, when, where — enough detail to repeat the study	Your process must satisfy all **four BRP conditions** (binary, independent, fixed p, fixed n)
Descriptive statistics	x, n, $\hat{p}$, and a simple chart	Describes the world you observed
Inferential statistics	Bar plot of the **null** binomial model (`dbinom` + `geom_col`), vertical line at your x, p-value from `pbinom`	Shows the world you imagine and how surprising your data is

Conclusion	One clear sentence tying p-value to the research question	Not "accept/reject H_0" — interpret evidence in context

> **!** One mistake to avoid early
>
> The **p-value is not** the probability that H_0 is true. It is the probability of data **at least this extreme**, *assuming* H_0 is true. You will say more about this in Read 2; keep the distinction in mind when you read the sample conclusion.

What you will do (different from the sample)

By the end of Module 1 you will:

1. Choose your **own** binomial process ($n = 25$), not ESP or other course examples.
2. Defend the four BRP conditions for **your** context.
3. Collect data, build the null model in **R**, compute a p-value, and write a **one-page** report with the same section headings as the sample.

The sample shows the **deliverable**. Read 1 and Read 2 teach the ideas; the demo lab and exercise lab teach the code; the Project 1 page has the full rubric and milestone checklist.

What to do next

1. Skim the sample PDF once (5–10 minutes).
2. Open Project 1 and note the due dates on the course path.
3. Start **Read 1** — probability language through the binomial pmf.

When you finish Read 1, take Quiz A (Def & Notation), then continue with Read 2.